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An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

**CS19611 - MOBILE APPLICATION DEVELOPMENT PROJECT
REPORT**

SIMPLE TODO LIST APP

Submitted by

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BONAFIDE CERTIFICATE

This project report titled "**SIMPLE TODO LIST APP**" is the bonafide work of **KEERTHANA S (220701124)**, who carried out the work under my supervision. Certified further that to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation based on which a degree or award was conferred earlier.

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CHAPTER 1

ABSTRACT

The “**SIMPLE TODO LIST**” application is a simple yet effective productivity tool developed using Android Studio with Kotlin as the primary programming language. The application allows users to efficiently manage daily tasks through a clean and user-friendly interface. Key functionalities include adding new tasks, editing existing entries, marking tasks as completed, and deleting tasks when no longer needed. The app utilizes local storage mechanisms to persist data, ensuring task lists remain intact across sessions. Designed with minimalism and usability in mind, this project demonstrates essential Android development concepts such as activity lifecycle, RecyclerView implementation, and Kotlin's concise syntax. The To-Do List app serves as a foundational project for understanding mobile app development and can be extended with features like cloud sync, notifications, and categorization in future iterations.

CHAPTER 2

INTRODUCTION

2.1 GENERAL

In today's fast-paced world, effective time and task management is critical. To-do list applications have become essential tools for organizing daily responsibilities.

With the growing number of mobile users, developing a mobile-based to-do app provides convenient access to productivity tools anytime, anywhere.

2.2 OBJECTIVE

The primary objective of this project is to build a simple and user-friendly to-do list application using Android Studio and Kotlin that enables users to:

- Add, edit, and delete tasks.
- Mark tasks as completed.
- Maintain task data even after closing the app (using persistent storage).

2.3 EXISTING SYSTEM

Existing to-do list applications, such as Google Tasks, Microsoft To Do, and others, provide a wide range of features but can be overwhelming for users who need a minimal, no-frills solution. Many such apps require cloud login or internet access, which may not be necessary for basic task management. Our proposed system simplifies this with offline support and local storage.

CHAPTER 3

LITERATURE SURVEY

Several mobile applications currently exist focusing on task management and personal productivity. Apps like **Google Tasks**, **Microsoft To Do**, and **Todoist** offer a variety of features for task organization, reminders, and cloud-based syncing. However, many of these existing systems often have:

- **Complex user interfaces** with feature overload, making them less suitable for users who require only basic functionality.
- **Mandatory sign-ins or cloud accounts**, increasing dependency on internet access.
- **Internet-based data storage**, which raises concerns regarding privacy and data access.
- **Subscription-based models** or premium features that limit usability in free versions.

Studies and user feedback highlight a growing demand for **simple, offline, and lightweight** to-do list apps that focus on core features like **adding, editing, and deleting tasks** without unnecessary complexity. Users prefer applications that are **minimalist**, easy to use, and do not require constant internet connectivity.

The proposed app aligns with these preferences, offering a streamlined task management experience with local storage using **Room Database** and a clean UI built with **Kotlin**.

CHAPTER 4

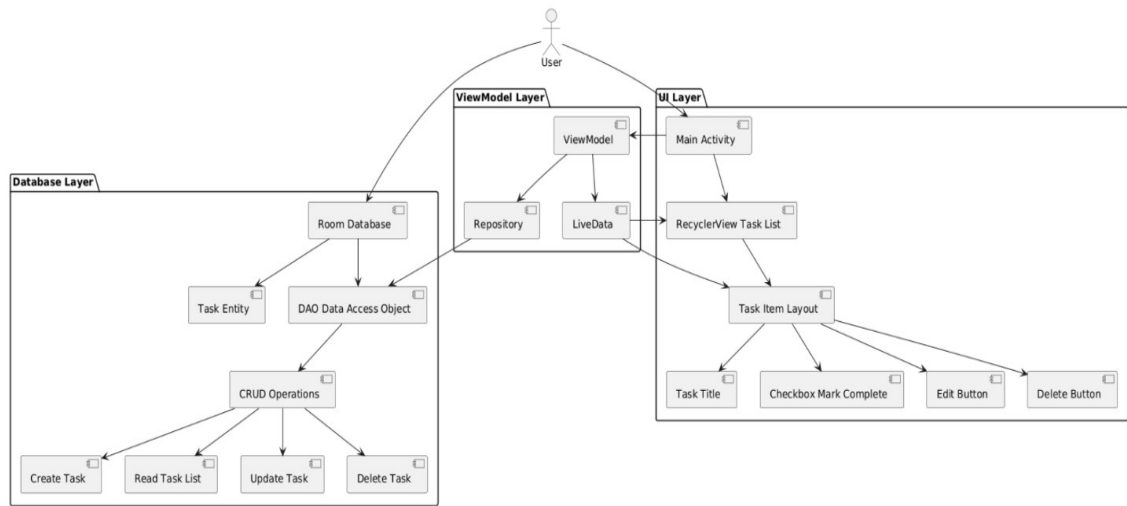
PROPOSED SYSTEM

4.1 SYSTEM OVERVIEW

The proposed system is a mobile-based to-do list application developed using Kotlin in Android Studio. It allows users to manage their daily tasks through functionalities like adding, updating, completing, and deleting tasks. Data persistence is achieved through the use of Room Database, ensuring that tasks remain available across app restarts. The app uses RecyclerView to display the task list dynamically.

4.2 SYSTEM ARCHITECTURE

- User opens the app and accesses the **To-Do Dashboard**.
- Tasks are displayed in a **RecyclerView** with:
 - Task Title
 - Checkbox (mark as complete)
 - Edit and Delete buttons
- All actions (add, update, delete, complete) are reflected instantly.
- Data is stored locally using **Room Database** to ensure persistence.



(Fig 3.1 System Architecture)

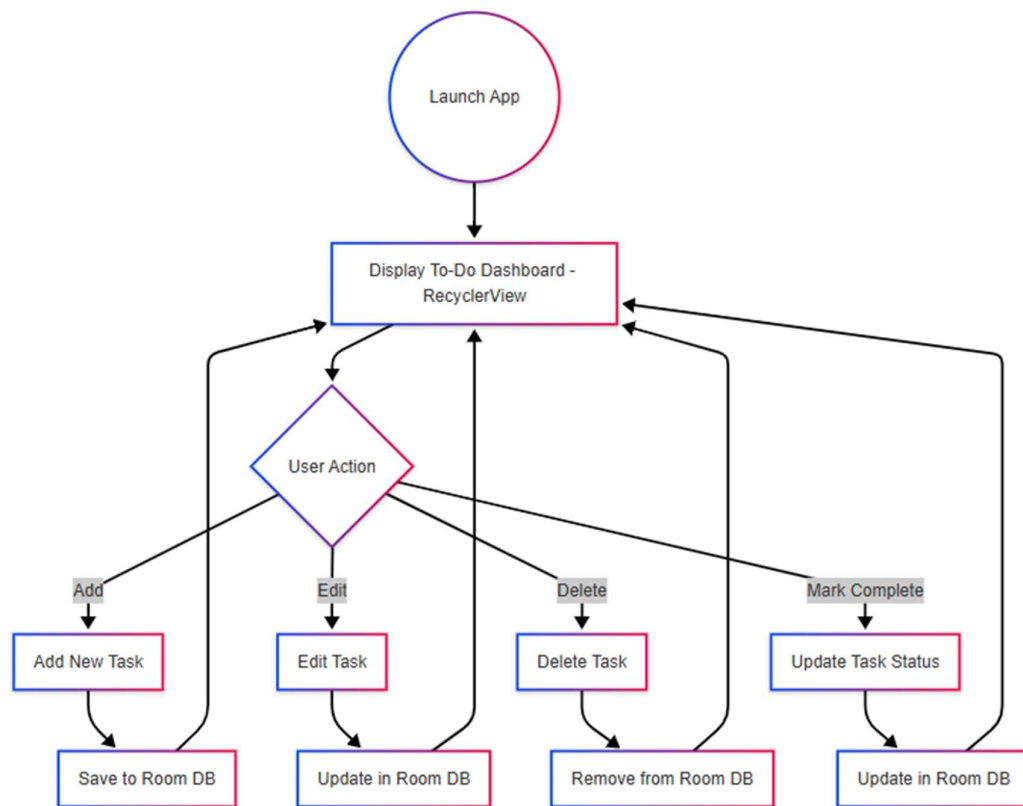
CHAPTER 5

MODULE DESCRIPTION

5.1 MODULES

- **Login** **Module:**
Allows access to the app through predefined static credentials. Ensures only authorized users can access and manage tasks.
- **Task Management** **Module:**
Enables users to add, view, update, complete, and delete tasks. Each task includes a title and completion status.
- **Room Database** **Module::**
Manages local data storage of tasks using **Room Database**. Ensures data persistence across sessions without internet connectivity..
- **UI/UX Module:**
Delivers a simple, intuitive interface following Material Design principles.
Uses **RecyclerView** for task lists with smooth interactions and a clean layout.

5.2 ACTIVITY DIAGRAM



(Fig 4.1 Activity Diagram)

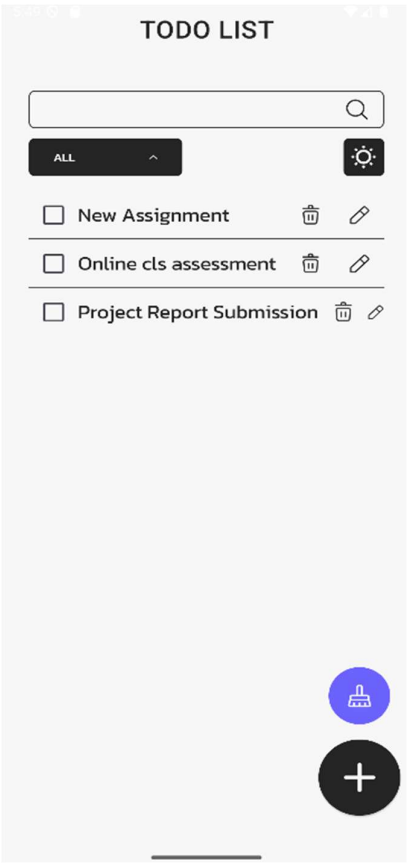
CHAPTER 6

IMPLEMENTATION AND RESULTS

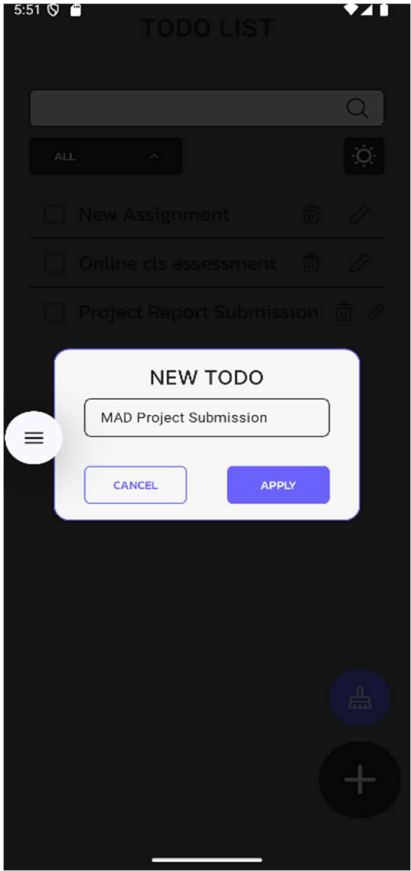
6.1 TOOLS USED

- Android Studio
- Kotlin
- XML for UI
- Room Database (for storing task)

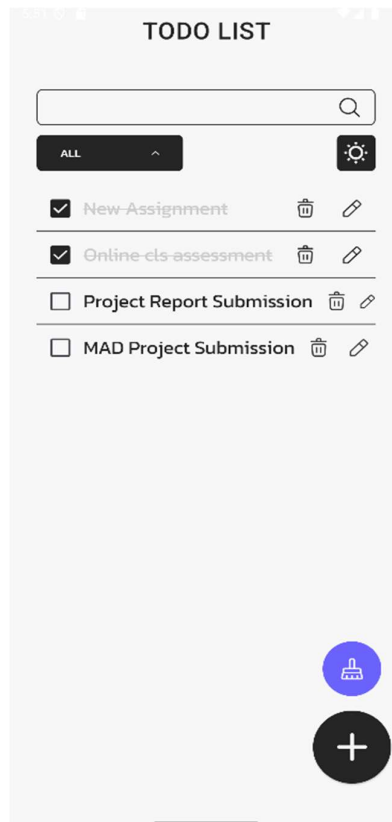
6.2 OUTPUT SCREENSHOTS



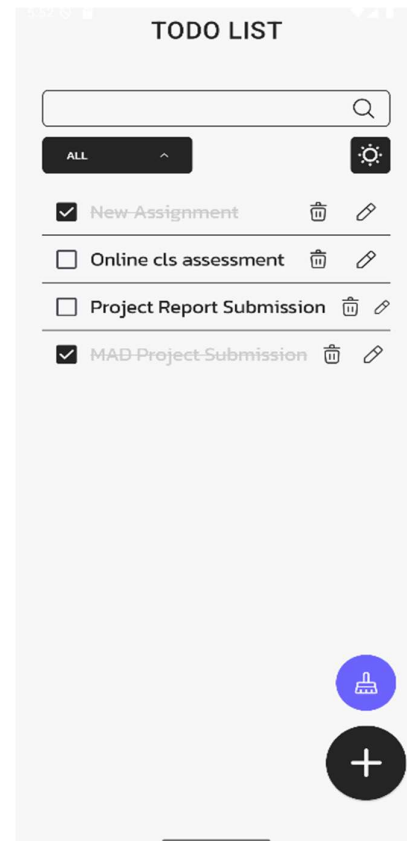
(Fig 6.1 Dashboard)



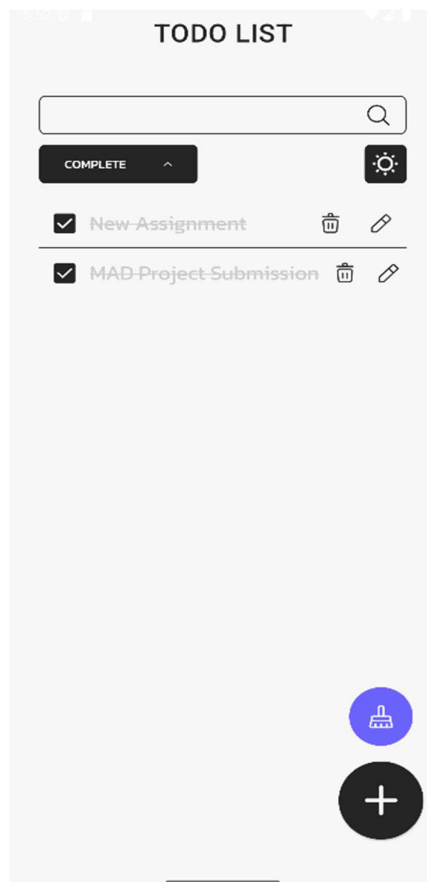
(Fig 6.2 Adding new task)



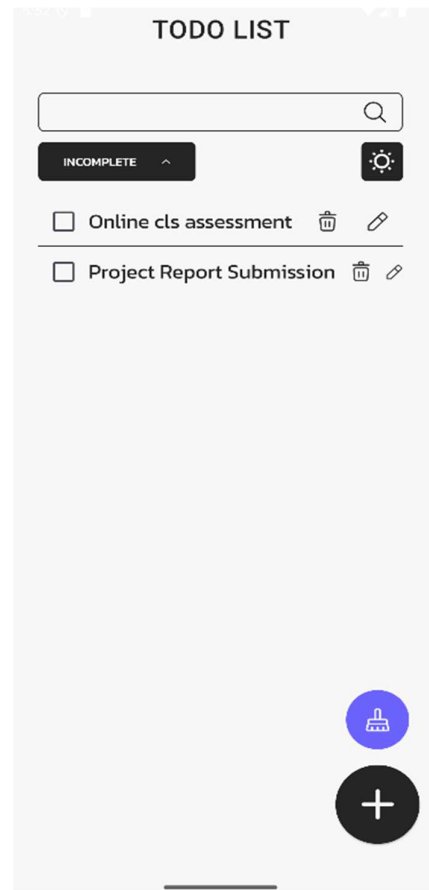
(Fig 6.3 Dashboard after adding new task)



(Fig 6.4 Checked the completed task)



(Fig 6.5 Displays completed task)



(Fig 6.6 Displays incomplete task)

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The To-Do List application offers a simple and efficient way for users to manage their daily tasks. Built using Kotlin and Android Studio with Room Database for persistent local storage, the app provides essential task management features such as adding, editing, deleting, and marking tasks as complete. Its clean UI and offline functionality make it an effective personal productivity tool.

7.2 FUTURE ENHANCEMENT

Integrate biometric login (fingerprint/face recognition) for secure access

- Add support for task reminders and due dates
- Implement category tagging (e.g., Personal, Work, Urgent)
- Enable cloud synchronization for multi-device access
- Include voice input for task creation
- Add a calendar view for better task visualization

REFERENCES

1. Android Developer Documentation
2. Kotlin Language Documentation
3. Room Database Guide – Android Jetpack
4. Material Design Guidelines (Google)